

Result: 70% 14 / 20	 100310 000216	Neptun ID: Draft papers: ☐ Name: _____
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The test consists of **5 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- | | a | b | c | d | e | f |
|-----|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1.) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 2.) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 3.) | ☐ | ☐ | ☐ | ☐ | | |
| 4.) | ☐ | ☐ | ☐ | ☐ | | |
| 5.) | (essay) | | | | | |

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

⊕

1.) Let $L^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language such that $Con = \{p, q\}$. Which string of symbols are membered in the set $Form$?

0 $(p \supset \neg q)$

2 $\neg \neg q$

(c) $(p + q)$

2 $\neg(\neg p \equiv \neg q)$

-1 $((p \supset \neg r) \equiv p)$

(f) $\neg p^*$

⊕

2.) Which statements are true for the formula below?

$$((\neg p \equiv q) \supset \neg(q \vee (\neg p \wedge q)))$$

(a) $(\neg p \wedge q)$ is a direct subformula

0 $((\neg p \equiv q) \supset \neg(q \vee (\neg p \wedge q)))$ is a subformula

(c) $(q \equiv \neg p)$ is a subformula

2 $(\neg p \wedge q)$ is a subformula

2 $(\neg p \equiv q)$ is a subformula

(f) $\neg(\neg p \wedge q)$ is a subformula

3.) Which interpretations are models of the following set of formulae?

$$\{ (q \supset (q \vee q)), \neg((q \equiv p) \supset p) \}$$

2 $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

(c) $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

4.) Which interpretations are models of the following set of formulae?

$$\{ (p \vee (q \supset q)), ((p \supset q) \supset \neg q) \}$$

2 $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

5.) When can we say that a set of formulae is unsatisfiable?

Score: 1 / 2.